THE HILBERT-POLYNOMIAL PIECE OF QUOT IS PROPER FOR PROPER X WITH AMPLE L

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1232-1244 of sources/stacks-moduli/pinned-original-source.tex; proof span is lines 1246-1267, the complete author proof (20 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 17 of queue_moduli.json (pb=1034, stmt 1232-1244, proof 1246-1267).

Editorial note. stacks-moduli wave-1: fresh adjudication accept (2026-10-14); pool per Chief order 08:5x (dpa superseded to lane-s9); moduli.tex 53 labels/48 with proof judged 25 accept / 23 reject / 5 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Editorial note. Author's 'some details omitted' covers the twist-invariance reparametrization bookkeeping; the conclusion rides on the named engine lemmas.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0DPB: sources/stacks-moduli/pinned-original-source.tex, source lines 1193–1201. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DP4: sources/stacks-moduli/pinned-original-source.tex, source lines 901–908. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01VS: sources/stacks-moduli/pinned-original-source.tex, source lines 1267–1267. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0D31: sources/stacks-moduli/pinned-original-source.tex, source lines 1267–1267. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:moduli.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 083R: sources/stacks-moduli/pinned-original-source.tex, source lines 1267–1267. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to B$ be a proper morphism of finite presentation of algebraic spaces. Let $\mathcal{F}$ be a finitely presented $\mathcal{O}_X$-module. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module ample on X/B, see Divisors on Spaces, Definition 0D31. The algebraic space $\mathrm{Quot}^P_{\mathcal{F}/X/B}$ parametrizing quotients of $\mathcal{F}$ having Hilbert polynomial P with respect to $\mathcal{L}$ is proper over B.*

Proof. The question is étale local over B, see Morphisms of Spaces, Lemma 083R. Thus we may assume B is an affine scheme. Then we can find a closed immersion $i : X \to \mathbf{P}^n_B$ such that $i^*\mathcal{O}_{\mathbf{P}^n_B}(1) \cong \mathcal{L}^{\otimes d}$ for some $d \geq 1$. See Morphisms, Lemma 01VS. Changing $\mathcal{L}$ into $\mathcal{L}^{\otimes d}$ and the numerical polynomial $P(t)$ into $P(dt)$ leaves $\mathrm{Quot}^P_{\mathcal{F}/X/B}$ unaffected; some details omitted. Hence we may assume $\mathcal{L} = i^*\mathcal{O}_{\mathbf{P}^n_B}(1)$. Then the isomorphism $\mathrm{Quot}_{\mathcal{F}/X/B} \to \mathrm{Quot}_{i_*\mathcal{F}/\mathbf{P}^n_B/B}$ of Lemma 0DP4 induces an isomorphism $\mathrm{Quot}^P_{\mathcal{F}/X/B} \cong \mathrm{Quot}^P_{i_*\mathcal{F}/\mathbf{P}^n_B/B}$. Since $\mathrm{Quot}^P_{i_*\mathcal{F}/\mathbf{P}^n_B/B}$ is proper over B by Lemma 0DPB we conclude. □